

BMJ Open Neoadjuvant adebrelimab combined with chemotherapy (cisplatin/carboplatin and etoposide) for limited-stage small-cell lung cancer: a study protocol of phase 2 trial (NIUS)

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To cite: Liu Y, Huang Q, Tian L, *et al.* Neoadjuvant adebrelimab combined with chemotherapy (cisplatin/carboplatin and etoposide) for limited-stage small-cell lung cancer: a study protocol of phase 2 trial (NIUS). *BMJ Open* 2025;**15**:e087302. doi:10.1136/bmjopen-2024-087302

► Prepublication history and additional supplemental material for this paper are available online. To view these files, please visit the journal online (<https://doi.org/10.1136/bmjopen-2024-087302>).

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Received 06 April 2024

Accepted 16 December 2024



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ABSTRACT

Introduction Small-cell lung cancer (SCLC) is a highly malignant neuroendocrine tumour, and concurrent chemoradiotherapy is the current recommended treatment for limited-stage SCLC. However, the overall survival (OS) of patients with SCLC remains poor. Therefore, improving the survival of patients with SCLC and benefitting more patients are urgent clinical requirements. Immunotherapy has achieved good efficacy in extensive-stage SCLC and non-SCLC. However, the use of neoadjuvant immunotherapy in limited-stage SCLC is unknown but promising. To this end, we plan to conduct a study on adebrelimab in combination with chemotherapy for the neoadjuvant treatment of limited-stage SCLC.

Methods and analysis This will be a single-arm, prospective study. The primary endpoint of this study will be a pathological complete remission rate. The secondary endpoints will be a major pathological response, objective response rate, event-free survival, OS, R0 resection rate, safety and potential predictive biomarker parameters related to efficacy in peripheral blood samples, including minimal residual disease, circulating tumour cells and cluster of differentiation (CD) 4/CD8. This study will enrol 28 patients.

Ethics and dissemination This study was approved by the Biomedical Ethics Review Committee of West China Hospital, Sichuan University, on 1 August 2023 (grant number 2023[987]). The version of this study protocol is V7.0. The patients have been enrolled since September 2023. This study will conclude in January 2026, and the results will be presented to peer-reviewed medical journals and international scientific conferences in 2027. This study will provide important data on whether neoadjuvant treatment with adebrelimab combined with chemotherapy is clinically beneficial in patients with limited-stage SCLC.

(Prospective study of neoadjuvant adebrelimab combined with chemotherapy in limited-stage SCLC; Chinese Clinical Trial Registry (ChiCTR), www.chictr.org.cn/; Number, ChiCTR2300074591.)

Trial registration number ChiCTR2300074591

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ Perioperative chemotherapy combined with immunotherapy, followed by surgery and postoperative immunomaintenance therapy was used in this study in limited-stage small-cell lung cancer.
- ⇒ Assays such as minimal residual disease were used to explore biomarkers to predict efficacy.
- ⇒ Although this is a multicentre study, the number of participants included was small owing to funding and time constraints.

INTRODUCTION

Small-cell lung cancer (SCLC) is a highly malignant neuroendocrine tumour that accounts for approximately 15% of all lung cancers.^{1 2} Compared with non-SCLC (NSCLC), SCLC has a higher proliferation index and faster doubling time and is prone to metastasis and chemoresistance. One-third of patients with SCLC are in a limited stage (tumour-node-metastasis stages I–III) at the time of the initial diagnosis.³ Currently, synchronous chemoradiotherapy is the recommended treatment option for limited-stage SCLC. Initially, SCLC responds to traditional chemotherapy, with an objective response rate (ORR) of approximately 60%, whereas drug resistance occurs quickly, with a median progression-free survival (PFS) time of only approximately 6 months and a median overall survival (OS) time of approximately 18 months, which makes the prognosis poor.^{4–6} Therefore, there is an urgent need to explore new therapeutic strategies for limited-stage SCLC. For example, the application of neoadjuvant therapy in limited-stage SCLC, in which tumours can be reduced by drug therapy, followed by surgical resection, thereby minimising the source

of drug-resistant tumours, may be a new option worth investigating.

The therapeutic value of surgery for limited-stage SCLC has been reported previously. In 2010, a large retrospective study analysed cases of limited-stage SCLC in the surveillance, epidemiology and end results database from 1988 to 2002.⁷ In total, 14179 patients were enrolled in this study, 863 of whom underwent surgical resection. The results showed that the 5-year survival rate of the surgical patients was 34.6%, with a median survival time of 28 months, and the 5-year survival rate of the nonsurgical patients was 9.9%, with a median survival time of 13 months, showing a significant difference between the two groups ($p<0.001$). Integrated multidisciplinary treatment (MDT), characterised by surgery combined with chemoradiotherapy, is significantly better than chemoradiotherapy alone in SCLC, and surgery has an important value in the treatment of SCLC.⁷⁻⁹ In 2023, the National Comprehensive Cancer Network and Chinese Society of Clinical Oncology guidelines stated that stages I–IIA SCLC could benefit from surgery and suggested that surgical treatment should be followed by combined adjuvant chemotherapy. However, for stages IIB–IIIA SCLC, the role of surgery is controversial, although some experts believe that surgery can benefit patients. For stage IIB SCLC, there remains a lack of adequate evidence to prove the survival benefit of surgery. Thus, studies on neoadjuvant therapy for limited-stage SCLC have rarely been reported. The limited level of surgical techniques and medical oncology treatments in the past may limit the application of surgery combined with medical treatment as a comprehensive treatment option for SCLC. In recent years, with the development of surgical techniques that reduce perioperative mortality, neoadjuvant immunotherapy has made significant breakthroughs in NSCLC, and CheckMate-816 has confirmed that neoadjuvant immunotherapy effectively improves the pathological complete remission rate (pCR) and event-free survival (EFS) of patients with NSCLC. Therefore, neoadjuvant immunotherapy has been approved by the US Food and Drug Administration and Chinese National Medical Products Administration as a standard treatment option for early- to mid-stage NSCLC. Simultaneously, especially immunotherapy, has brought about the possibility of neoadjuvant therapy for SCLC.

Three phase III clinical studies, IMpower133, CASPIAN and CAPSTONE-1, have successfully demonstrated that programmed death-ligand 1 monoclonal antibody immunotherapy in combination with chemotherapy improves OS and PFS times in the first-line treatment of extensive-stage SCLC.¹⁰⁻¹⁴ Among them, the results of the CAPSTONE-1 study showed that adebrelimab in combination with chemotherapy for the first-line treatment of extensive-stage SCLC resulted in a median OS time of 15.3 months (vs 12.8 months with chemotherapy), a 2-year survival rate of 31.3% (vs 17.2% with chemotherapy) and a significant reduction in the risk of disease progression of patients by up to 33% compared with that

of the chemotherapy group.¹⁴ These findings demonstrate that immunotherapy combined with chemotherapy has a better therapeutic effect on SCLC than conventional chemotherapy alone. In the basic study, our team used multiple immunofluorescence methods to distinguish cold or hot tumour immune microenvironment in the preliminary study and found that ‘hot’ microenvironment was presented in 20 of 36 patients with limited-stage SCLC, whereas that was observed in only one of nine patients with extensive-stage SCLC. This suggests that limited-stage SCLC may have a more favourable immune microenvironment than extensive-stage SCLC and that immunotherapy may be equally effective in limited-stage SCLC. The use of immunotherapy in combination with chemotherapy in the neoadjuvant and adjuvant treatments of limited-stage SCLC will likely result in greater clearance of subclinical foci of tumours and a breakthrough in therapeutic efficacy and survival benefits.

This new concept has recently been put into clinical practice in a single-arm, multicentre study of neoadjuvant/transform treatment with atezolizumab in combination with chemotherapy in patients with resectable SCLC in a collaboration between Professor Xiaolong Yan’s team and Professor Zhe Liu’s team.¹⁵ The primary methodology of the study was that patients with untreated limited-stage SCLC received three cycles of neoadjuvant/transform therapy with a regimen of atezolizumab combined with etoposide and platinum-based chemotherapy, followed by surgery. The results showed that the pCR and major pathological response (MPR) for the intention-to-analyse cohort were 47.1% (8/17) and 70.6% (12/17), respectively, and the ORR for the protocol-compliant cohort was 100%. The median OS time for patients with pCR and the median EFS time for those who underwent surgery has not yet been achieved, whereas the median OS time for patients without pCR was 18.2 months, and the median EFS time for nonsurgical patients was 9.5 months. These results suggest that immunotherapy combined with chemotherapy shows more significant efficacy as a neoadjuvant treatment for limited-stage SCLC compared with chemotherapy alone. In addition, we once reported a case of stage IIB SCLC with significant tumour regression after immunotherapy followed by combination chemotherapy, after which the patient underwent lung cancer resection. A pathological complete response (CR) was observed, and DFS time exceeded 3 years without recurrence or metastasis.¹⁶ This evidence suggests that the efficacy and survival benefits of neoadjuvant immunotherapy combined with chemotherapy in limited-stage SCLC warrant further exploration.

In summary, the evidence suggests neoadjuvant treatment with immunotherapy combined with chemotherapy followed by radical surgical resection for SCLC is of high scientific value and clinical significance. Therefore, we propose to conduct a study on the neoadjuvant treatment of limited-stage SCLC with adebrelimab-combined chemotherapy (cisplatin/carboplatin and etoposide). Neoadjuvant

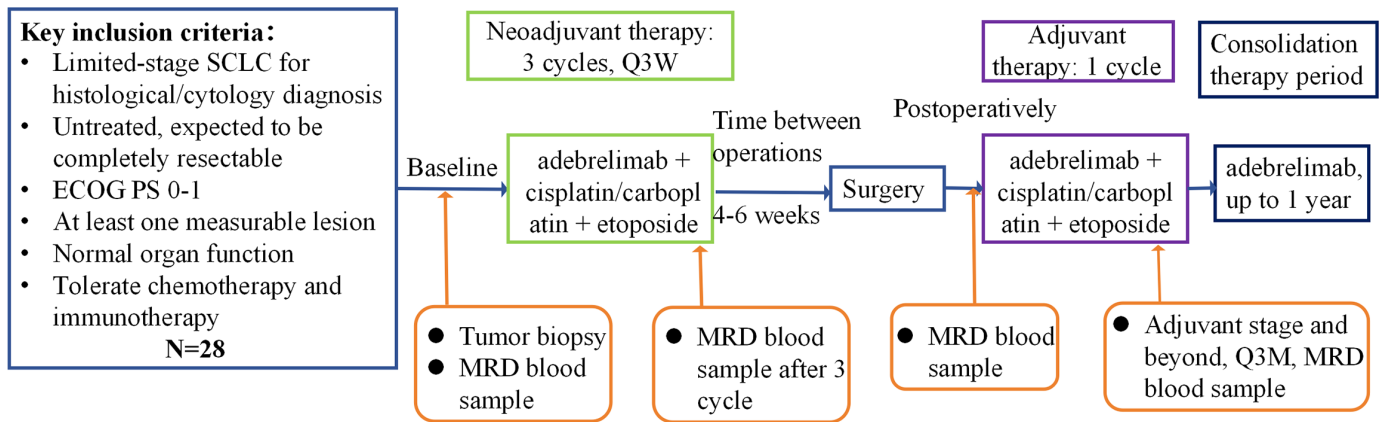


Figure 1 Protocol scheme of this study. ECOG, Eastern Cooperative Oncology Group; MRD, minimal residual disease; PS, performance status. NOTE: This study will use Simon's two-stage design. Eleven participants will be included in the first phase. If more than one participant is satisfied with pCR, the study will proceed to the next phase, and conversely, the study will be terminated. A total of 28 participants will be enrolled in the two phases, and more than four participants will need to be satisfied to develop pCR; otherwise, the study will be terminated.

immunotherapy for limited-stage SCLC is unknown but promising, and our study provides important research data for the exploration of new SCLC treatments.

AIM OF THE STUDY

1. Main objective: to evaluate the antitumour efficacy of neoadjuvant adebrelimab in combination with chemotherapy (cisplatin/carboplatin and etoposide) for limited-stage SCLC.
2. Secondary objective: to evaluate the safety of neoadjuvant adebrelimab in combination with chemotherapy (cisplatin/carboplatin and etoposide) for limited-stage SCLC.
3. Exploratory research objective: to explore potential predictive biomarkers related to efficacy in peripheral blood samples, such as minimal residual disease (MRD).

METHODS AND ANALYSIS

Study design

This will be a single-arm, multicentre and prospective study conducted at the Lung Cancer Centre of West China Hospital, 363 Hospital and Daping Hospital. The protocol is illustrated in figure 1. Professor Yan Zhang will be the chief investigator. The study duration will be 3 years. Patients with SCLC will be recruited from September 2023 to August 2024. The expected treatment duration for the last patient will be from September 2024 to January 2026. If the actual duration of the study changes, it is not a violation of the protocol.

Eligibility criteria of the participants

The participants will voluntarily participate in this study, sign an informed consent form and cooperate with the follow-up procedures.

Inclusion criteria

1. Limited-stage SCLC confirmed by imaging and histological/cytopathological examination (resectable and potentially resectable SCLC), with patients with resectable SCLC referring to T1–2, N0 patients who are physically fit and willing to undergo surgery and patients with potentially resectable SCLC referring to patients with tumours invading major vessels, trachea or other important irremovable organs, or although there is lymph node metastasis, but only N1 or solitary N2 lymph node metastasis, patients who may undergo a reduction operation after immunotherapy combined with chemotherapy (based on the American Joint Committee on Cancer V. 8)
2. Potential to undergo radical surgery based on NCCN and CSCO guidelines (R0 resection).
3. CT or MRI scanning ≤ 28 days before the first study drug administration, with at least one measurable tumour lesion (meeting the Response Evaluation Criteria in Solid Tumours (RECIST) V. 1.1 standards)
4. No prior antitumour therapy
5. Age between 18 and 75 years, regardless of sex
6. Surgery planned after completion of neoadjuvant therapy
7. Preoperative organ function showing no contraindications for surgery
8. Eastern Cooperative Oncology Group performance status of 0–1 points
9. Key organ functions meeting the following requirements (no blood components and cell growth factors are allowed within 2 weeks before starting the study treatment):
Complete blood count must satisfy the following: (a) absolute neutrophil count $\geq 1.5 \times 10^9/\text{L}$, (b) haemoglobin $\geq 9\text{g/dL}$, (c) platelet count $\geq 100 \times 10^9/\text{L}$ and (d) serum albumin $\geq 2.8\text{g/dL}$.
Biochemistry must comply with the following: (a) total bilirubin $\leq 1.5 \times \text{ULN}$, (b) alanine aminotransferase,

aspartate aminotransferase $\leq 2.5 \times \text{UILN}$, (c) serum creatinine $\leq 1.5 \times \text{ULN}$ and endogenous creatinine clearance rate $\geq 50 \text{ mL/min}$ (Cockcroft–Gault formula) and (d) normal thyroid function.

10. Expected survival period ≥ 3 months.

11. Ability to take atezolizumab combined with chemotherapy.

12. Female participants capable of childbirth undergoing urine or serum pregnancy tests within 72 hours before the first administration of the study drug with negative results and willing to take effective contraceptive methods during the trial and within 3 months after the last administration of adebrelimab and male participants whose partners are women of childbearing age using effective contraceptive methods during the trial and within 3 months after the last administration of atezolizumab.

13. Good compliance and cooperation with the follow-up.

Exclusion criteria

1. Known allergy to the study drugs or any of its excipients or severe allergic reaction to other monoclonal antibodies.
2. Combined with other tumours.
3. Any active autoimmune disease or history of autoimmune diseases, such as interstitial pneumonia, uveitis, enteritis, hepatitis, pituitary inflammation, vasculitis, myocarditis, nephritis, hyperthyroidism and hypothyroidism (participants may be enrolled in this study when they are on hormone replacement therapy and the indices are normalised)
4. Poor physical condition.
5. Hypertension that cannot be well controlled using antihypertensive drugs (systolic blood pressure $\geq 140 \text{ mmHg}$ or diastolic blood pressure $\geq 90 \text{ mmHg}$).
6. Positivity for hepatitis B surface antigen and hepatitis B virus DNA test values exceeding the upper limit of normal (1000 copy numbers/mL or 500 IU/mL), hepatitis C virus (HCV) positivity (HCV RNA or HCV antibody testing indicates acute or chronic infection), known HIV-positive history or acquired immune deficiency syndrome history.
7. Taking steroids (prednisone efficacy dose $> 10 \text{ mg/day}$) or other immunosuppressants for systemic treatment within 14 days before the first use of the study drug (in the absence of an active autoimmune disease, inhaled or topical steroids and adrenocorticotrophic hormone replacement therapy at doses equivalent to prednisone at $> 10 \text{ mg/day}$ will be permitted).
8. Severe infections within 4 weeks prior to first-time medication use, including but not limited to infectious complications requiring hospitalisation, bacteraemia and severe pneumonia and active infection, with cases of lymphatic spread due to lung cancer not excluded.
9. Grade II or higher myocardial ischaemia or myocardial infarction and poorly controlled arrhythmias (including QTc intervals $\geq 450 \text{ ms}$ in men and ≥ 470

ms in women) and grades III–IV cardiac insufficiency according to the New York Heart Association criteria or cardiac ultrasonography showing a left ventricular ejection fraction $< 50\%$

10. History of psychotropic substance abuse incapable of quitting or having mental disorders
11. Other factors that may have forced the termination of the involvement in the study prematurely (eg, other serious diseases (mental illness) requiring concomitant treatment, severely abnormal laboratory test results, social or family matters and difficulties in long-term follow-up)
12. Any other circumstances unfit for inclusion in this study

Discontinuation criteria

Treatment termination criteria

1. Withdrawal of informed consent and request for withdrawal
2. Disease progression as determined by the investigator
3. Inability to tolerate the side effects of immunotherapy and chemotherapy
4. Other circumstances that the investigator deems necessary to withdraw from this study

Patient withdrawal criteria

1. Withdrawal of informed consent and refusal of further follow-up
2. Missed visits
3. Death
4. Termination of this study
5. Other circumstances deemed necessary by the investigator to warrant withdrawal from this study.

Study termination criteria

1. Unintended, meaningful or unacceptable risks to participants.
2. Ineffectiveness of the study drug/trial treatment or continuation of the trial that is pointless.
3. The declarant deciding to terminate the study for reasons, such as a significant lag in participant enrollment and frequent protocol deviations.

PATIENT BACKGROUND DATA

After obtaining informed consent from the registered clinical investigators of each institute (informed consent form written in Chinese is shown in the online supplemental materials), baseline data (sex, age, performance status, comorbidities, smoking history, allergies, vital signs, electrocardiography findings, blood and urine examination results, pulmonary function test results, chest and abdominal CT and brain MRI findings) will be collected to determine eligibility. These data will also be collected before resection.

TREATMENT METHODS

The neoadjuvant regimen will be adebrelimab (1200 mg, D1, Q3W) combined with cisplatin (75 mg/m^2) or

carboplatin (AUC=5, Q3W) plus etoposide (100 mg/m²) (Q3W) for three cycles. The specific time of chemotherapy drug administration will depend on the actual operational requirements of the centre.

After the preoperative therapy, radical surgery (lobectomy and mediastinal lymph node dissection) will be performed in all patients with resectable or potentially operable SCLC. Participants with resectable SCLC refer to those who are fit for surgery and willing to operate, such as patients with stage T1-2N0 physical status. Participants with potentially resectable SCLC refer to those whose tumour invades the roots of the large blood vessels, trachea or other unresectable vital organs; who have lymph node metastasis, but only N1 or single N2 lymph node metastasis; and who have the possibility of downstaging surgery after immunotherapy combined with chemotherapy. In addition, these patients should have good pulmonary functional reserve and, according to the RECIST version 1.1 criteria, resectable patients should have tumour efficacy assessed as CR, partial response (PR) or stable disease. Patients with potentially resectable tumours and tumour efficacy will be assessed as achieving CR or PR. Surgery will be performed within 4–6 weeks of the preoperative therapy completion. Moreover, prophylactic brain irradiation and palliative thoracic radiotherapy may be administered after surgery according to the site's protocols, with the exception of curative thoracic radiotherapy for treating or eliminating residual disease.

The adjuvant therapy regimen will be adebrelimab (1200 mg, D1, Q3W) combined with cisplatin (75 mg/m²) or carboplatin (AUC=5) plus etoposide (100 mg/m², Q3W) for one cycle. The specific time of chemotherapeutic drug administration will depend on the actual operational requirements of the centre.

The consolidated treatment regimen will include adebrelimab (1200 mg, D1, Q3W). If the participant's disease progresses or develops an intolerable toxic reaction, if the medications received by the participant will no longer be clinically effective or if the participant will receive another antitumour therapy, withdraw informed consent or die, consolidation therapy will be discontinued. Consolidation therapy should not exceed a maximum of 1 year.

OUTCOMES

The primary outcome is pathological complete response (pCR).

The secondary outcomes are MPR, ORR, EFS, OS and R0 resection rate.

The potential predictive biomarker parameters related to efficacy in peripheral blood samples, including MRD, circulating tumour cells, and cluster of differentiation CD4/CD8.

FOLLOW-UP AND ASSESSMENT

The safety follow-up period during the study period will be 90 days (± 7 days) after the last antitumour therapy or prior to initiation of new antitumour therapy, whichever came first, every 30 days (± 7 days), and can be conducted by telephone. All adverse events will be assessed using the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI-CTCAE) version 5.0. The adverse events will be recorded during the follow-up period.

Survival follow-up will be performed immediately after the safety follow-up, and participants will be followed up for survival every 3 months (± 28 days), with telephone interviews (or clinical follow-ups) with the participants themselves, their family members or their local physicians until the onset of disease progression or disease recurrence, or loss to follow-up, which lasted for 1 year. See [table 1](#) for the specific items.

Statistical analyses

In this study, the data will be summarised using descriptive statistics according to the following general principles, unless otherwise stated. Measurement data will be summarised using the mean, SD, median, maximum, and minimum values. Count data will be summarised using frequency and percentage, and time-to-event data will be summarised using the Kaplan–Meier estimation of survival and plotting of survival curves.

The primary endpoint for neoadjuvant therapy will be the pCR rate. The number of patients and percentage of participants achieving pCR will be summarised, and the 95% confidence intervals for the rate will be calculated using the Clopper–Pearson method.

Adverse events will be evaluated using the NCI-CTCAE version 5.0. The analysis for adverse events will be based on a treatment-wide analysis set. Data analysis will include, but is not limited to, calculating the incidence of adverse reactions in groups, listing the frequency and number of adverse reactions using the system, and calculating percentages and detailed listings of cases of various adverse events.

All statistical analyses will be performed using the SPSS 22.0 statistical analysis software. All statistical tests will be performed using two-sided tests, where a $p\text{-value} \leq 0.05$ will be considered statistically significant for the differences tested, and a 95% confidence level will be used for the confidence intervals.

SAMPLE SIZE CALCULATION

Simon's two-stage design will be used. Referring to previous studies of neoadjuvant chemotherapy for SCLC, the P0: pCR rate will be 9.4%, and we assume that the P1: pCR rate will be 27%, and the parameters in this study will be unilateral $\alpha=0.1$ and $1-\beta=0.8$. According to the optimisation principle, 11 cases will be included in the first stage. If more than one case of pCR will occur during the trial, the study will proceed to the next stage; otherwise, it will be terminated. A total of 28 patients will be

Table 1 Testing items and testing time points

Project	Screening period		Neoadjuvant therapy period		Pre-surgical examination		Surgery		Adjuvant therapy and consolidated treatment period	
	D-28~D-1	C1-C3	D1	D1	Surgery will be performed within 4–6 weeks of the preoperative therapy completion		C4-C17		D1	Survival follow-up
Sign informed consent	✓									
Demographic data collection	✓									
Tumour diagnosis	✓									
Past medical history	✓									
History of previous antineoplastic therapy	✓									
Physical examination	✓				✓					
Vital signs	✓		✓		✓			✓		
ECOG score	✓		✓		✓			✓	✓	
Complete blood count	✓ (within 7 days)		✓		✓			✓		
Comprehensive metabolic panel	✓ (within 7 days)		✓		✓			✓		
Coagulation function	✓ (within 7 days)		✓		✓			✓		
Routine urinalysis	✓ (within 7 days)		✓		✓			✓		
Stool analysis	✓ (within 7 days)		✓		✓			✓		
Thyroid function test	✓ (within 7 days)		✓		✓			✓		
HIV/HBV/HCV virus testing	✓									
Pregnancy test	✓ (within 7 days)									
12-lead ECG	✓ (within 7 days)		✓		✓			✓		
Echocardiography	✓				✓					
Imaging (chest)	✓				✓ (within 7 days before surgery)			Assessment every four cycles		
Imaging (parts of the body other than the chest)	✓				✓					
Pathological response rate assessment							✓ (Postoperative)			
Peripheral blood sample acquisition	✓				✓ (3 weeks after the last dose of neoadjuvant therapy to 7 days before surgery)		✓ (3–4 weeks after surgery)	Acquisition every four cycles		
Adverse events	✓		✓		✓		✓	✓	✓	
Concomitant medications/concomitant therapies			✓		✓		✓	✓	✓	✓
Time to disease progression									✓	✓
Follow-up anti-tumour therapy								✓	✓	✓

enrolled in both stages. If four or more cases will show pCR, this study will be continued; otherwise, this study will be terminated. This study will include 28 patients.

PATIENTS AND PUBLIC INVOLVEMENT

Neither patients nor members of the public were directly involved in this study.

INFORMED CONSENT

All participants will be informed about the purpose of this study and the intervention plan of the study and alternative treatment options, benefits and possible risks for the participants. Subsequently, the patients will sign a consent form. All the participants will be provided sufficient time to consider their participation in this study. Patients participating in the study will be allowed to withdraw at any stage without restrictions.

PARTICIPATING INSTITUTIONS

Lung Cancer Centre of West China Hospital, 363 Hospital and Daping Hospital will be the participating institutions.

DATA STORAGE AND SECURITY DATA

Data will be collected via an electronic data capture system hosted by West China Hospital with a secure web interface. In accordance with the regulations of clinical research at the West China Hospital, we will establish a data safety monitoring programme according to the level of risk. All adverse events will be recorded in detail, handled appropriately and followed up until they resolve or stabilise. Serious and unintended adverse events will be reported to the ethical review committee. The competent authority, sponsor and drug regulatory authority, in a timely manner according to the regulations, will regularly conduct a cumulative review of all adverse events and convene a meeting of the investigators to assess the risks and benefits of the study, as necessary.

MONITORING

Changes made to the protocol will be communicated to the ethics committee and will be included in the clinical trial registry.

This study will use appropriate personnel with degrees in medicine, pharmacy or related professions who have undergone the necessary training and are familiar with good clinical practice and relevant regulations as monitors of this clinical trial. They will monitor the conduct of the trial and verify the data to ensure that the rights and interests of the participants in the clinical trial are safeguarded; that the data recorded and reported in the trial are accurate, complete and error-free; and that the trial is conducted in accordance with the approved protocol and relevant regulations.

DISCUSSION

The treatment of SCLC has entered the era of immunotherapy. Data from the ADRIATIC trial has shown that adjuvant therapy with durvalumab after standard concurrent platinum-based chemoradiotherapy led to significantly longer OS and PFS than placebo among patients with limited-stage SCLC.¹⁷ However, it is worth noting that in clinical practice, we have observed that many patients with limited-stage SCLC do not tolerate concurrent chemoradiotherapy well. How these patients with limited-stage SCLC can derive maximum benefit from immunotherapy requires further investigation. Based on the design and results of the ADRIATIC trial study, we speculate that in the era of immunotherapy, a treatment regimen combining immunotherapy, chemotherapy and local therapy has significant application potential in the treatment of limited-stage SCLC.

In addition to radiotherapy, radical surgery is also an effective method for local treatment of lung cancer. Indeed, the perioperative holistic treatment model, ranging from preoperative neoadjuvant therapy to postoperative adjuvant therapy for lung cancer, has received widespread attention. Several studies, including the Checkmate 816 study, have demonstrated that this combination regimen can effectively improve the pCR and EFS of patients with NSCLC. However, whether immunotherapy combined with chemotherapy can achieve the same significant neoadjuvant therapeutic effects in limited-stage SCLC remains to be explored.

Based on our experience with treated patients and clinical research data from other researchers, we have found that neoadjuvant immunotherapy is a promising approach worth further study in limited-stage SCLC. Therefore, we have decided to conduct this prospective study, aiming to further develop treatment methods for limited-stage SCLC, thereby providing more treatment options for patients with limited-stage SCLC in different disease states.

We design a neoadjuvant treatment study of adefrelimab combined with chemotherapy (cisplatin/carboplatin and etoposide) in patients with limited-stage SCLC. Compared with the previous similar study, our study is novel and standardised in terms of treatment protocols. We use new immunotherapy agents, enrol a larger number of subjects and explore new biomarkers of treatment efficacy. We will conduct exploratory observations of MRD levels and changes in immune cell types and numbers before and after neoadjuvant therapy to identify patients with limited-stage SCLC who are suitable for neoadjuvant treatment with immunotherapy in combination with chemotherapy. The limitations of this study include its small sample size. The main purpose of this study is to assess the pCR rates. Based on the extrapolation of pCR data for the neoadjuvant treatment of SCLC reported in the previous literature, we believe that this number of cases is sufficient to fulfil the main purpose. This will be a multicentre clinical trial that can reduce study bias. The main research centre of this study will be

the Lung Cancer Centre of West China Hospital, which is an MDT unit that integrates surgery and medical treatment. It enables the highly qualitative management of patients throughout the entire process.

In summary, if the current study meets its primary endpoint, it will demonstrate that adebrelimab in combination with chemotherapy (cisplatin/carboplatin and etoposide) for limited-stage SCLC increases efficacy and safety. The results of this study have a high scientific value and clinical significance for the treatment of limited-stage SCLC.

Ethics and dissemination

This study follows the relevant provisions of the Declaration of Helsinki of the World Medical Assembly and the Measures for Ethical Review of Biomedical Research Involving Human Beings of the National Health and Family Planning Commission of the People's Republic of China. The trial protocol was approved by the Biomedical Ethics Review Committee of West China Hospital.

This study provides important information on whether neoadjuvant treatment with adebrelimab in combination with chemotherapy (cisplatin/carboplatin and etoposide) is clinically beneficial for limited-stage SCLC. For participants, the results of this study will be announced by their physicians if favourable pathological responses are achieved for included patients.

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Contributors YZ was responsible for the conceptualization and methodology. Formal analysis was done by YZ and YL. Investigation was performed by YL, QH, LT, XZ, GC, HX, ML, DZ and YZ. YZ was responsible for the resources. Writing-original draft preparation was done by YZ, YL and QH. Writing-review and editing was performed by YZ. YZ was responsible for the supervision. Project administration was done by YZ, HX and ML. YZ is the guarantor in this study.

Funding This work was supported by the Key R&D Projects of the Health Commission of Sichuan Province (No. 24LCYJZD09) and the 1·3·5 project for disciplines of excellence-Clinical Research Incubation Project, West China Hospital, Sichuan University (Grant No. 2019HXFH062)

Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Patient consent for publication Not applicable.

Provenance and peer review Not commissioned; externally peer reviewed.

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